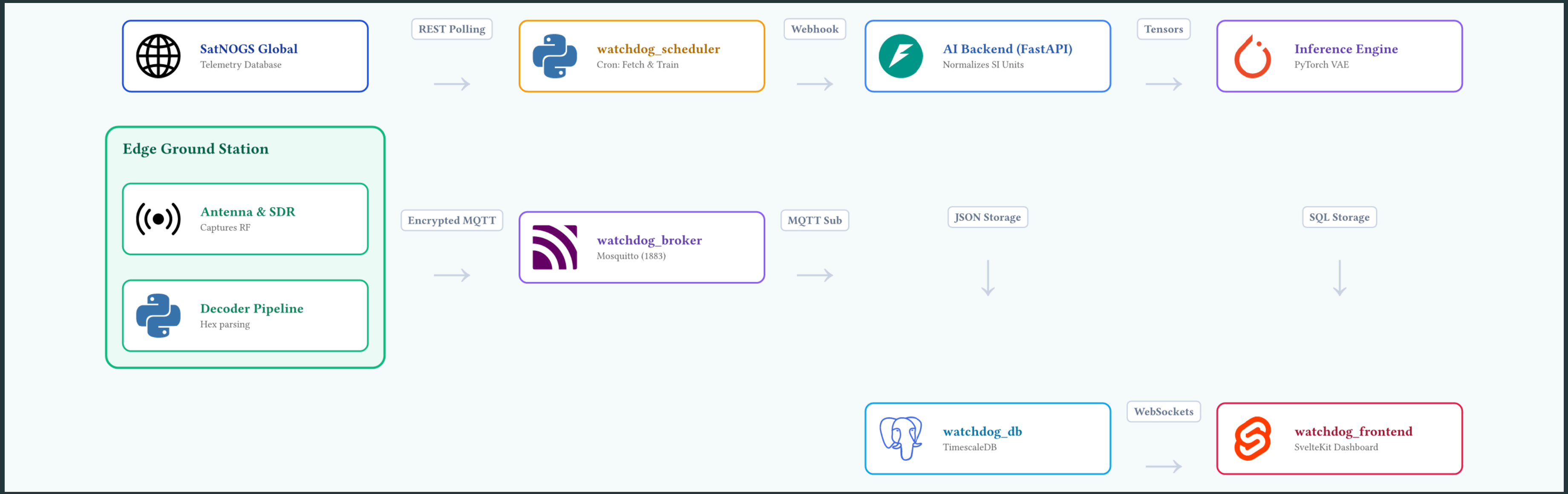
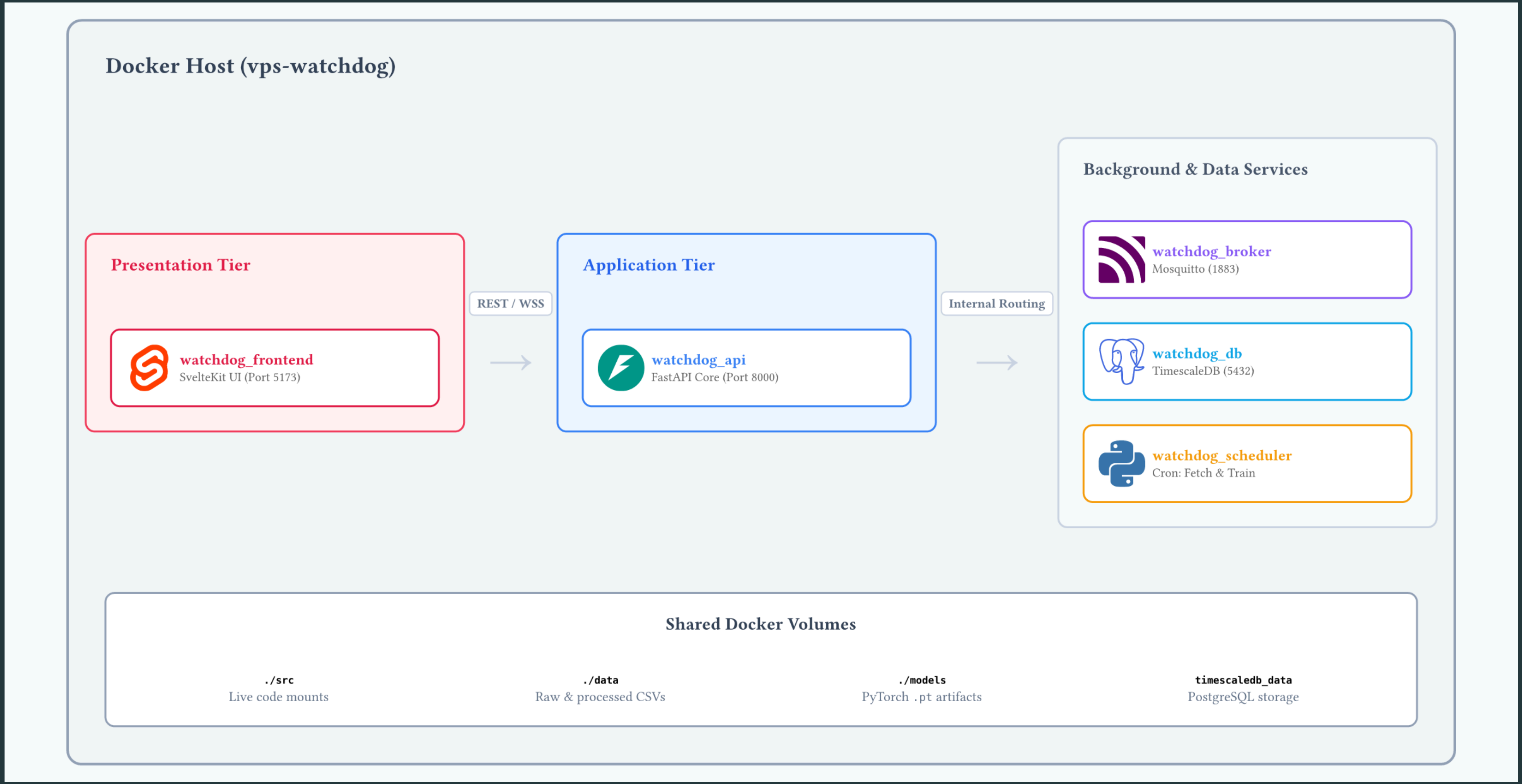
UWE-4 System Architecture

Docker Topologies & Edge Integration

Architecture & Integration


SatNOGS Global
Telemetry Database
REST Polling
watchdog_scheduler
Cron: Fetch & Train
Webhook
AI Backend (FastAPI)
Normalizes SI Units
Tensors
Inference Engine
PyTorch VAE
Edge Ground Station
Antenna & SDR
Captures RF
Decoder Pipeline
Hex parsing
Encrypted MQTT
watchdog_broker
Mosquitto (1883)
MQTT Sub
JSON Storage
SQL Storage
watchdog_db
TimescaleDB
WebSockets
watchdog_frontend
SvelteKit Dashboard


Docker Host (vps-watchdog)
Presentation Tier
watchdog_frontend
SvelteKit UI (Port 5173)
REST / WSS
Application Tier
watchdog_api
FastAPI Core (Port 8000)
Internal Routing
Background & Data Services
watchdog_broker
Mosquitto (1883)
watchdog_db
TimescaleDB (5432)
watchdog_scheduler
Cron: Fetch & Train
Shared Docker Volumes
./src
Live code mounts
./data
Raw & processed CSVs
./models
PyTorch .pt artifacts
timescaledb_data
PostgreSQL storage

Implementation Logic

The Shared Core Pipeline

We use the "Shared Core" (`src/gr_sat/telemetry.py`) which acts as the universal adapter between raw bits and our AI model.

Pipeline Steps:

1. **Ingest:** Read raw hex frames (SatNOGS/SDR).
2. **Parse:** Kaitai Structs via `satnogs-decoders`.
3. **Normalize:** Convert binary fields to SI Units (V, A, °C).
4. **Validate:** Check against physical limits.

Current Status:

- Core Logic: **Operational (Shared Core)**
- Decoder Ecosystem: **satnogs-decoders**
- Coverage: **1 production decoder in-repo (UWE-4)**
- Online Runtime: **Implemented (FastAPI + TimescaleDB + SvelteKit UI)**

Validation & Edge Benchmarking

Because real anomaly data is rare, we validate the model using **Synthetic Fault Injection** on clean data.

Accuracy (Recall / FPR)

1. **Panel Failure:** High temp, but negative current.
2. **Thermal Runaway:** Inject large positive battery-temp steps.
3. **Sensor Stuck:** Historical notebook experiment, not in current shipped benchmark.

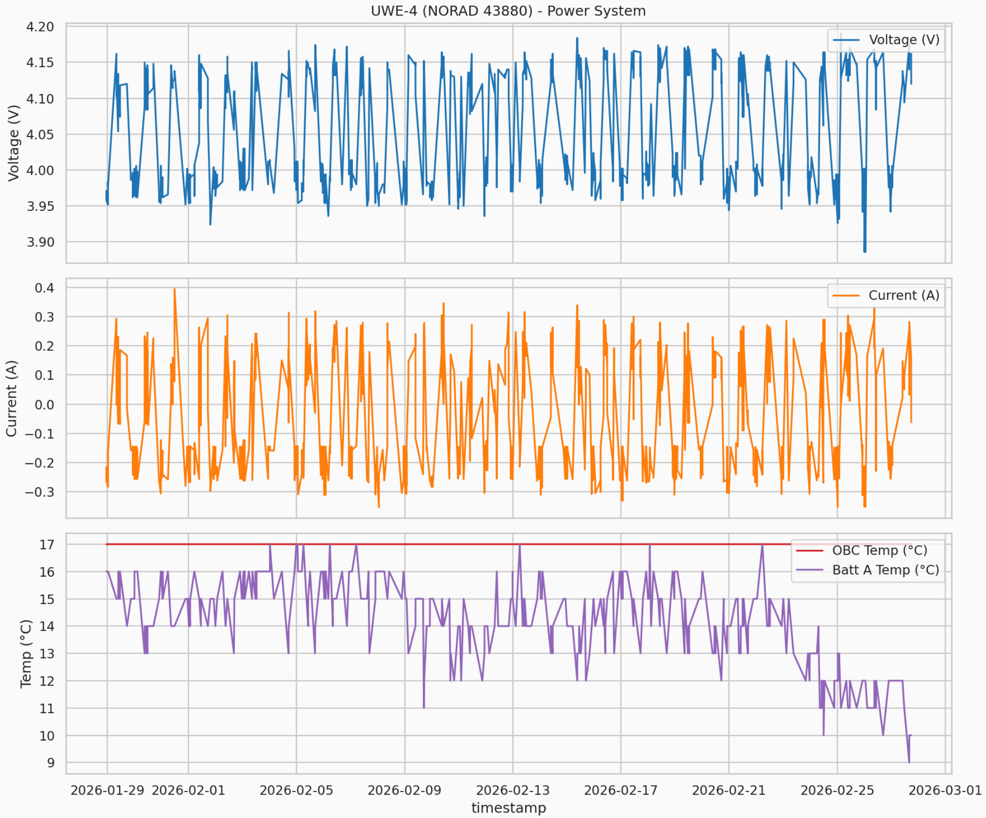
Edge Targets (Online Runtime)

Integrated with Dockerized Backend / MQTT Broker.

- **Latency:** target < 10 ms per frame.
- **Memory:** target < 5 MB model footprint.

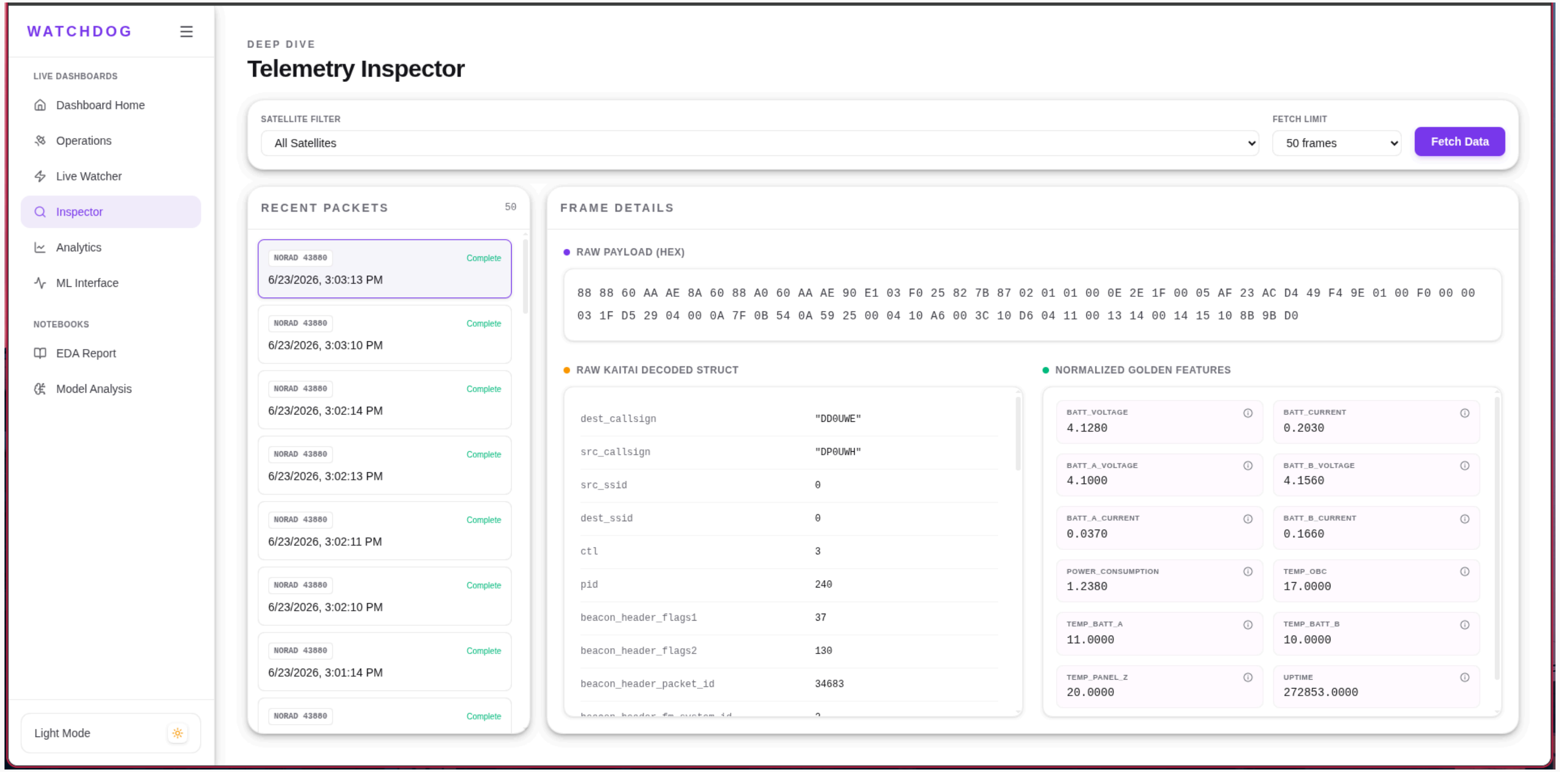
Data Verification (UWE-4)

Telemetry extraction from UWE-4 (NORAD 43880). Perfect mapping from `satnogs-decoders` to our Golden Features (Volts, Amps, °C).



The Inspector Tool

We built an interactive debugger (`telemetry_inspector`) to verify decoders against real historical data.



Summary of Phase 3

1. **Targeting:** Locked onto **UWE-4** (43880) as our primary target.
2. **Pipeline:** Processing is operational, successfully generating structurally clean ML inputs.
3. **Data Strategy:** Fetching 180+ days of data to capture seasonal variations.